CATS$^+$

Competishun Advance

Training Sessions

(Lecture - II)

Que

The value of $\sum_{r=1}^{\infty} \frac{(4r+5)5^{-r}}{r(5r+5)}$ is :

(A) $\frac{1}{5}$ ✓ (B) $\frac{2}{5}$ (C) $\frac{1}{25}$ (D) $\frac{2}{25}$

$$T_r = V_r - V_{r-1}$$

$$\sum_{r=1}^{\infty} \frac{(4r+5)\cdot 5^{-r}}{r(5r+5)} = \sum_{r=1}^{\infty} \frac{4r+5}{r(r+1)\,5^{r+1}}$$

$$= \sum_{r=1}^{\infty} \Big(\underbrace{\frac{1}{r\cdot 5^r}}_{V_r} - \underbrace{\frac{1}{(r+1)5^{r+1}}}_{V_{r+1}} \Big)$$

$r=1$ $\quad$ $r \to \infty$

$$= \frac{1}{1\cdot 5} - 0 = \frac{1}{5}$$

$\boxed{A}$

Consider

$$\frac{1}{r\cdot 5^r} - \frac{1}{(r+1)\cdot 5^{r+1}}$$

$$= \frac{5\cdot(r+1) - r}{r\cdot(r+1)\cdot 5^{r+1}}$$

$$= \frac{4r+5}{r(r+1)\cdot 5^{r+1}}$$

Paragraph for Question Nos. 1 to 2

There are two sets A and B each of which consists of three numbers in A.P. whose sum is 15. D and d are their respective common differences such that $D - d = 1$, $D > 0$. If $\frac{p}{q} = \frac{7}{8}$ where p and q are the product of the numbers in those sets A and B respectively

$a-d, \cancel{a}, a+d$

$3a = 15$

$a = 5$

1. Sum of the product of the numbers in set A taken two at a time is :

(A) 51 (B) 71 (C) 74 (D) 86

2. Sum of the product of the numbers in set B taken two at a time is :

(A) 52 (B) 54 (C) 64 (D) 74

$D = d + 1$

$A \equiv \{5-D, 5, 5+D\}$

$B \equiv \{5-d, 5, 5-d\}$

$$\frac{p}{q} = \frac{7}{8} = \frac{(5-D)\cdot \cancel{5} \cdot (5+D)}{(5-d)\cdot \cancel{5} \cdot (5+d)} = \frac{25-D^2}{25-d^2}$$

$$\cancel{7 \times 25} - 7d^2 = \cancel{25 \times 8} - 8 \cdot D^2$$

$$8D^2 - 7d^2 = 25$$

$$8D^2 - 7d^2 = 25 \qquad D = d+1$$

$$8(d+1)^2 - 7d^2 = 25$$

$$d^2 + 16d - 17 = 0.$$

$$d = 1, -17. \qquad D = 2 \text{ or } -16$$

$$A = \{3, 5, 7\}$$

$$15 + 21 + 35 = 71$$

$$B = \{4, 5, 6\}$$

$$20 + 24 + 30 = 74$$

Que The number of solutions of the equation

$$z^2 = \bar{z} \cdot 2^{1-|z|} \text{ is :}$$

(A) 0 (B) 2 ~~(C) 4~~ (D) None of these

Soln

$$z^2 = \bar{z} \cdot 2^{1-|z|}$$

$|\cdot|$

$$\Rightarrow |z|^2 = |z| \cdot 2^{1-|z|}$$

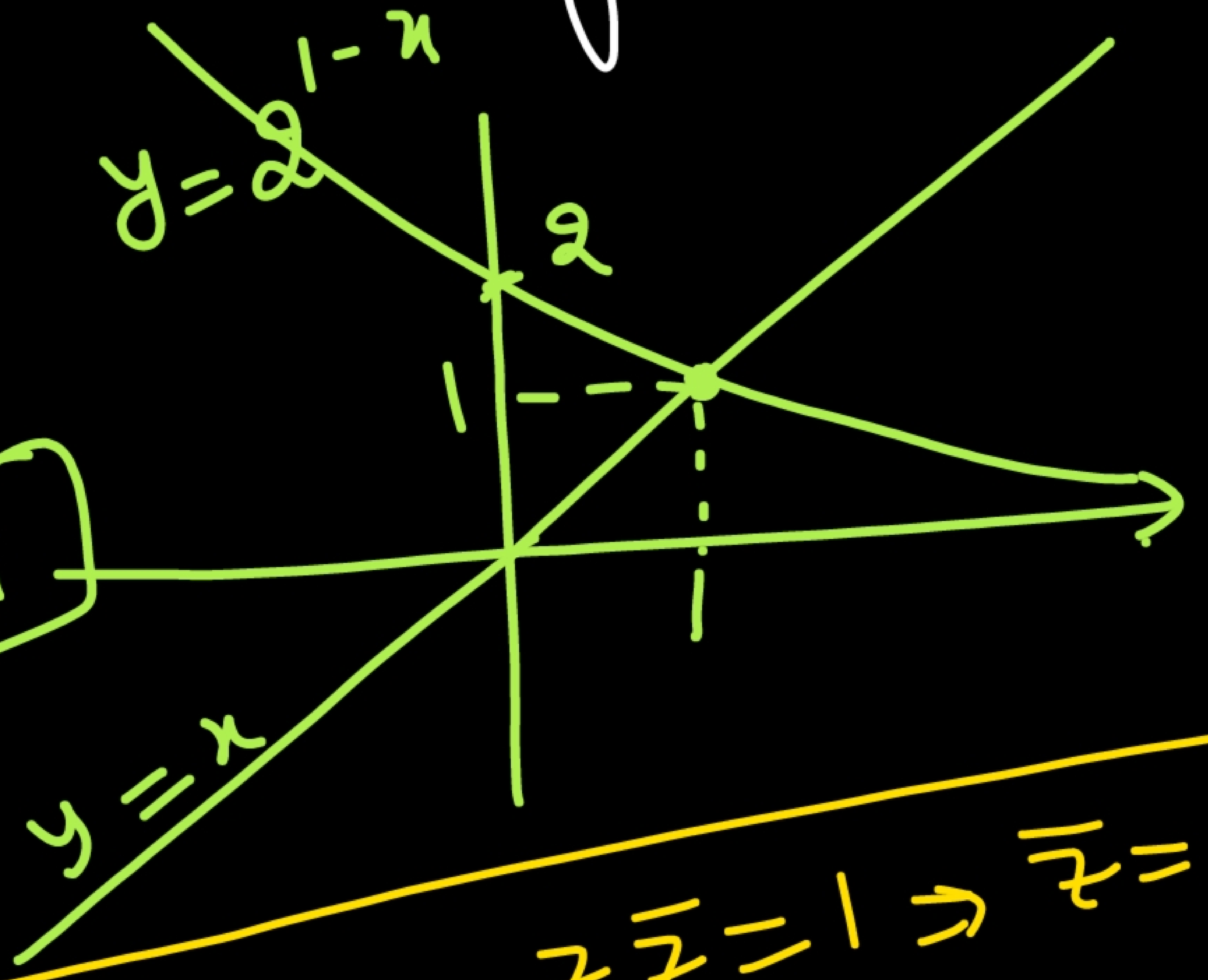
$|z| = 0$ OR $|z| = 2^{1-|z|}$

$\hookrightarrow \boxed{z=0}$ ✓

Let $|z| = x$

$$x = 2^{1-x}$$

$\hookrightarrow \boxed{x=1}$



$\hookrightarrow |z| = 1.$

$|z| = 1 \Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$

G. Eqn. $z^2 = \bar{z} \cdot 2^{1-|z|}$

$$z^2 = \frac{1}{z} \cdot 2^0 = \frac{1}{z}$$

$$z^3 = 1$$

$$z^3 = 1 \rightarrow z = 1, \omega, \omega^2$$

$$z = 0$$

$$\boxed{z = 0, 1, \omega, \omega^2}$$

$$z^2 = \bar{z} \cdot 2^{1-|z|}$$

$z = \omega$

$$\omega^2 = \omega^2 \, \cancel{2^{1-1}}$$

$z = \omega^2$

$$\omega = \omega \cdot 2^{1-1} = \omega$$

$$\bar{\omega} = \omega^2$$

$$\overline{\omega^2} = \omega$$

Que The sequence $a_1, a_2, a_3, \ldots$ satisfy $a_1 = 1$, $a_2 = 2$

and $a_{n+2} = \dfrac{2}{a_{n+1}} + a_n$, $n = 1, 2, 3, \ldots$

then the value of $\dfrac{2^{2019}}{2021} \cdot a_{2022}$ is

A $\quad {}^{2020}C_{1010}$

B $\quad {}^{2021}C_{1016}$

C $\quad {}^{2021}C_{1015}$

D $\quad$ None of these

$$a_{n+2} = \frac{2}{a_{n+1}} + a_n$$

$$a_{n+2} \cdot a_{n+1} = 2 + a_{n+1} \cdot a_n$$

$$n \to n-1 \quad \begin{cases} a_{n+2} \cdot a_{n+1} - \cancel{a_{n+1} \cdot a_n} = 2 \\ \cancel{a_{n+1} \cdot a_n} - \cancel{a_n \cdot a_{n-1}} = 2 \\ \cancel{a_n \cdot a_{n-1}} - \cancel{a_{n-1} a_{n-2}} = 2 \\ \vdots \\ + \; \cancel{a_3 a_2} - a_2 a_1 = 2 \end{cases}$$

$$a_{n+2} \cdot a_{n+1} - a_2 a_1 = 2n$$

$$a_2 \to 2, \quad a_1 \to 1$$

$$a_{n+2} \cdot a_{n+1} = 2 + 2n = 2(n+1)$$

$$n \Rightarrow n-1 \quad \begin{cases} a_{n+2} \cdot a_{n+1} = 2(n+1) \\ a_{n+1} \cdot a_n = 2n \end{cases}$$

$$\div$$

$$\frac{a_{n+2}}{a_n} = \frac{n+1}{n}$$

$$\boxed{a_{n+2} = \left(\frac{n+1}{n}\right) \cdot a_n}$$

$$a_{n+2} = \left(\frac{n+1}{n}\right)\cdot a_n$$

Now

$$a_{2022} = \frac{2021}{2020}\, a_{2020} = \frac{2021}{2020}\cdot\frac{2019}{2018}\cdot a_{2018} = \frac{2021}{2020}\cdot\frac{2019}{2018}\cdot\frac{2017}{2016}\cdot a_{2016}$$

$$a_{2022} = \frac{2021}{2020}\cdot\frac{2019}{2018}\cdot\frac{2017}{2016}\cdots\cdots\cdot\frac{3}{2}\cdot \cancel{a_2}^{\,2}$$

$$= \frac{(2021)(2019)(2017)\cdots 3}{2^{1010}\left((1010)(1009)\cdots 3\cdot 2\cdot 1\right)}\cdot 2$$

$$= \frac{(2021)(2020)(2019)(2018)\cdots\cdots 3\cdot 2}{2^{1010}\,(1010)\left((2020)(2018)\cdots\cdots 2\right)}\cdot 2$$

$$2 \qquad 2 \qquad\qquad 2$$

$$a_{2022} = \frac{\lfloor\underline{2021}}{2^{1010} \cdot 2^{1010} \cdot \lfloor\underline{1010}\left((1010)(1009)(1008)\cdots 2.1\right)} \cdot 2$$

$$= \frac{\lfloor\underline{2021}}{2^{2019} \cdot \lfloor\underline{1010}\;\lfloor\underline{1010}} = \frac{2021}{2^{2019}} \cdot \underbrace{\frac{\lfloor\underline{2020}}{\lfloor\underline{1010}\;\lfloor\underline{1010}}}_{\to\ {}^{2020}C_{1010}}$$

$$\frac{2^{2019}}{2021} \cdot a_{2022} = \boxed{{}^{2020}C_{1010}}.$$

One

If 2 sin x sin y + 3 cos y + 6 cos x sin y = 7, then the greatest value of $\tan^2 x + 2\tan^2 y$ is a, also $\sum_{k=0}^{\infty} \frac{2^k}{7^{2^k}+1} = \frac{b}{c}$ (b and c are coprime). Which of the following alternative(s) is/are correct?

$a = 9$
$b = 1$
$c = 6$

(A) a + b + c = 15 ✗
(B) a + b − c = 5 ✗
(C) a−b + c=14 ✓
(D) b + c − a = −2 ✓

$2\sin(x)\cdot\sin(y) + 3\cos(y) + 6\cos(x)\cdot\sin(y) = 7$

$a = \left(\tan^2(x) + 2\tan^2(y)\right)_{max}$

$2\sin(x)\cdot\sin(y) + 3\cos(y) + 6\cos(x)\cdot\sin(y)$

$\Rightarrow \underbrace{2(\sin(x) + 3\cos(x))}_{a}\cdot\sin(y) + \underbrace{3}_{b}\cdot\cos(y)$

$L.H.S.]_{max} = \sqrt{a^2+b^2} = \sqrt{4(s+3c)^2 + 9}$

$\geq \sqrt{4(\sqrt{10})^2 + 9}$

$\Rightarrow L.H.S.]_{max} = 7 = R.H.S. = 7$

for maximum value of L.H.S.

$\sin(x) + 3\cos(x) = \sqrt{10}$

$2\sqrt{10}\sin(y) + 3\cos(y) = 7$

$\sin(x) + 3\cos(x) = \sqrt{10}$.

$\downarrow$

max. value $= \underline{\underline{\sqrt{10}}}$

$= f(x)$ say

$f(x))_{max.} = \sqrt{10}$

for which $f'(x) = 0$.

$\cos(x) - 3\sin(x) = 0$

$\tan(x) = 1/3$.

$2\sqrt{10}\ \underbrace{\sin(y) + 3\cos(y)}_{\text{max}\ 7} = 7$

$= g(y) \Rightarrow g(y))_{max.} = 7$

$\Downarrow$

$g'(y) = 0$

$2\sqrt{10}\cos(y) - 3\sin(y) = 0$

$\tan(y) = \frac{2\sqrt{10}}{3}$

$\tan^2(x) + 2\tan^2(y) = \frac{1}{9} + 2\cdot\frac{4\times 10}{9} = \frac{81}{9} = 9 = a$

$\boxed{a = 9}$ Ans

$$\sum_{k=0}^{\infty} \frac{2^k}{7^{2^k}+1} = \frac{b}{c}$$

$$(a^2-b^2) = (a-b)(a+b)$$

$$x^2-1 \equiv (x-1)(x+1)$$

$$(a^2+1)(x^2+1) = x^4-1$$

$x = 7$

$$\text{L.H.S.} = \frac{1}{x+1} + \frac{2}{x^2+1} + \frac{4}{x^4+1} + \frac{8}{x^8+1} + \cdots \infty$$

$$= \frac{1}{x-1} - \frac{1}{x-1} + \frac{1}{x+1} + \frac{2}{x^2+1} + \frac{4}{x^4+1} + \cdots$$

$$-\frac{2}{x^2-1} \Rightarrow -\frac{4}{x^4-1} + \rightarrow -\frac{8}{x^8-1} + \rightarrow \cdots$$

$$= \frac{1}{x-1} - \lim_{n\to\infty} \frac{2^n}{x^{2^n}-1} = \frac{1}{x-1} = \frac{1}{7-1} = \frac{1}{6} = \frac{b}{c} \Rightarrow b = 1,\ c = 6.$$

$$2^n = t \longrightarrow \infty$$

$$\lim_{t \to \infty} \frac{t}{x^t - 1} = 0$$

$$\left(\frac{\infty}{\infty}\right) \quad \text{L.H.L.} \quad \frac{1}{x^t \cdot \ln(x)} = \frac{1}{\infty}$$

Que Evaluate

$$\cos a \cos 2a \cos 3a \ldots\ldots\ldots \cos 999a, \text{ where } a = \frac{2\pi}{1999}$$

let

$$C = \cos(a) \cdot \cos(2a) \cdot \cos(3a) \text{ - - - - - - } \cos(999a)$$

$$S(\theta) \cdot C(\theta) = \frac{S(2\theta)}{2}$$

$$S = \sin(a) \cdot \sin(2a)\ \sin(3a) \text{ - - - - } \sin(999a)$$

$$C \cdot S = \frac{\sin(2a) \cdot \sin(4a) \cdot \sin(6a) \text{ - - - - } \sin(1998a)}{2^{999}}$$

$$C \cdot S = \sin\left(\frac{4\pi}{1999}\right) \sin\left(\frac{8\pi}{1999}\right) \cdot \sin\left(\frac{12\pi}{1999}\right) \text{ - - - - - - } \sin\left(\frac{1998 \cdot 2\pi}{1999}\right)$$

$$C.S. = \frac{1}{2^{999}}\left(\sin(2a)\cdot\sin(4a)\cdot\sin(6a) - - - - - - \sin(1998a)\right)$$

$$= \frac{1}{2^{999}}\left(\sin(2a)\sin(4a) - - - \sin(998a).\sin(1000a).\sin(1002a) - - - \sin(1998a)\right)$$

$$a = \frac{2\pi}{1999}$$

$$= \sin\left(\frac{2000\pi}{1999}\right) = -\sin\left(\frac{\pi}{1999}\right) = -\sin\left(\frac{1998\pi}{1999}\right)$$

$$= -\sin(999a)$$

$$= \sin\left(\frac{2004\pi}{1999}\right) = -\sin\left(\frac{5\pi}{1999}\right)$$

$$= -\sin\left(\frac{1994\pi}{1999}\right)$$

$$= -\sin(997a)$$

$$\underline{1000a} = (1999 - 999)a$$

$$= 1999a - 999a.$$

$$= 2\pi - 999a.$$

$$\sin(1000a) = -\sin(999a)$$

$$1002a = (1999 - 997)a = 2\pi - 997a.$$

$$S \cdot C = \frac{1}{2^{999}}\left(S(2a) \cdot S(4a) \cdots S(998a) \cdot (-S(999a))(-S(997a)) \cdots (-S(a)) \right)$$

$$\cancel{S} \cdot C = \frac{1}{2^{999}}\left(\cancel{\sin(a) \cdot \sin(2a) \cdot \sin(3a) \cdots \sin(999a)} \right)$$

$$\Rightarrow C = \boxed{\frac{1}{2^{999}}} \text{ Ans}$$